Task 1 - Understand the Problem and Identify the Data

C Programming Assignment - Rural Electricity Billing System

Objective

Build a C program that calculates a household electricity bill using meter readings.

Inputs and outputs

Inputs: household number, category, previous reading and current reading. Outputs: units consumed, energy charge, fixed charge and total bill.

Formula

Units consumed = current reading minus previous reading. Total bill = energy charge plus fixed charge. Calculate energy charges slab by slab.

Data and validation

Use integer paise for money, wide integers for meter readings and totals, and enum for customer category. Reject malformed input, negative readings and current readings below previous readings unless a confirmed meter rollover is handled.

Complexity

For S tariff slabs, calculation is O(S) time and O(1) extra space. With a fixed number of slabs, it is O(1).

Important: illustrative tariff rates are used because the screenshot does not show the complete official tariff table. Replace them with your instructor's prescribed rates.